

Microservices Exercise 3: API Gateway Routing

Exercises on Microservices with Spring Boot 3.0

1. Build a User and Order Management System

Problem: Create two microservices:

- ? User Service to manage users.

- ? Order Service to manage orders placed by users.

Requirements:

- ? Use REST APIs.

- ? Communicate between services using WebClient (Spring WebFlux) or OpenFeign.

- ? Store data in MySQL or PostgreSQL.

2. Inventory Management System with Service Discovery

Problem: Create:

- ? Product Service: Manage products and stock.

- ? Inventory Service: Track stock levels for each product.

Requirements:

- ? Use Spring Cloud Netflix Eureka for service discovery.

- ? Implement centralized configuration using Spring Cloud Config Server.

3. Implement an API Gateway

Problem: Create an API Gateway to route requests to:

- ? Customer Service

- ? Billing Service

Requirements:

- ? Use Spring Cloud Gateway.

- ? Implement rate limiting, caching, and path rewriting.

4. Resilient Microservices with Circuit Breaker

Problem: A Payment Service calls a slow third-party API.

Requirements:

- ? Implement Circuit Breaker and fallback logic using Resilience4j.

- ? Log and monitor fallback events.